Kidney International, Vol. 58 (2000), pp. 762–769

Long-term comparison of losartan and enalapril on kidney function in hypertensive type 2 diabetics with early nephropathy

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Long-term comparison of losartan and enalapril on kidney function in hypertensive type 2 diabetics with early nephropathy.

Background. The objectives of this study were to compare the effects of the angiotensin II receptor blocker, losartan, to those of the angiotensin-converting enzyme inhibitor, enalapril, on albuminuria and renal function in relationship to clinic and ambulatory blood pressure (ABP) in hypertensive type 2 diabetic subjects with early nephropathy. The tolerability of these agents and their effect on the metabolic profile were also evaluated.

Methods. The study was a one-year prospective, double-blind trial with losartan and enalapril administered alone or in combination with hydrochlorothiazide and other antihypertensive agents. ABP and renal and biochemical parameters were measured at baseline and after 12, 28, and 52 weeks of active treatment. Ninety-two hypertensive type 2 diabetics with early nephropathy completed the study.

Results. Both losartan and enalapril administered alone or in combination with other agents induced significant reductions in sitting clinic ($P < 0.05$) and ABP ($P < 0.002$) without a statistical difference between groups. Geometric means for urinary albumin excretion (UAE) decreased significantly ($P < 0.001$) in patients treated with losartan from 64.1 to 41.5 μg/min and in those treated with enalapril from 73.9 to 33.5 μg/min after 52 weeks of therapy. A significant relationship ($P < 0.05$) between changes in systolic and diastolic ABP and the decrease in UAE at 52 weeks was seen in both groups. The decline in glomerular filtration rate (GFR) was stabilized at the end of therapy and was identical in both treatment groups. Treatment with enalapril was associated with a significantly higher incidence of cough ($P = 0.006$) and a rise in serum uric acid ($P = 0.002$) compared with losartan.

Conclusions. Our results indicate that a one-year course of antihypertensive therapy with either losartan or enalapril significantly reduces UAE in hypertensive type 2 diabetic patients with early nephropathy. The reduction in UAE with each treatment is similarly related to decrements in ABP. In addition, the rate of decline in GFR is similar in both treatment groups.

Key words: hypertension, NIDDM, early nephropathy, uric acid, antihypertensive therapy.

Received for publication July 28, 1999
and in revised form February 7, 2000
Accepted for publication March 8, 2000

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Several rigorous longitudinal clinical studies have demonstrated that increased urinary albumin excretion (UAE) predicts clinical proteinuria and increased mortality. In addition, increased UAE also predicts increased morbidity, especially hypertension and cardiovascular disease, in non–insulin-dependent diabetes mellitus (NIDDM) or type 2 diabetic patients [1–4].

Various studies have shown that treatment with angiotensin-converting enzyme (ACE) inhibitors lowers blood pressure (BP) and reduces both albuminuria and the rate of decline in the glomerular filtration rate (GFR) in hypertensive type 2 diabetics with or without early nephropathy [5–9]. Recently, a new class of drugs that selectively inhibits the renin-angiotensin-aldosterone system (RAAS) by specifically targeting the angiotensin II type 1 (AT_1) receptor has been developed for the treatment of hypertension. Specific antagonists of the AT_1 receptor have been shown to be efficacious antihypertensive agents and are associated with a significantly lower frequency of the unwanted adverse effects of ACE inhibitors, such as cough [10, 11]. Moreover, these agents reduce proteinuria and maintain GFR as effectively as ACE inhibitors in hypertensive patients with renal disease [12, 13]. A recent three-month study with losartan or enalapril performed in patients with essential hypertension and normal renal function resulted in a similar and significant reduction of UAE [14]. To date, however, no study has compared the long-term effects of BP reduction with AT_1 antagonists to those of ACE inhibitors on

UAE and renal function in hypertensive type 1 or type 2 diabetics.

Consequently, in the present study, we compared the effects of 12 months of treatment with losartan, the first available AT_1 antagonist, to those of the ACE inhibitor enalapril on UAE and the relationship to clinic and ambulatory BP reduction in hypertensive type 2 diabetics with persistent increased UAE. A secondary objective was to monitor GFR during antihypertensive therapy. In addition, we evaluated the tolerability of these agents and their effects on the metabolic profile.

METHODS

This multicenter, double-blind study was conducted at eight clinical centers in Canada. Institutional review board approval was obtained from each center, and all patients gave their informed consent. All procedures performed in this study were conducted in accordance with institutional guidelines.

Patient selection

Male and female outpatients with type 2 diabetes mellitus diagnosed at 30 years of age or later, mild to moderate essential hypertension [sitting diastolic BP (SIDBP) 90 to 115 mm Hg], and early nephropathy characterized by a UAE rate 20 to 350 μg/min without evidence of urinary tract infection were screened for enrollment.

Exclusion criteria included evidence or suspicion of renovascular disease, history of malignant hypertension, systolic BP > 210 mm Hg, cerebrovascular accident in the previous 12 months or current transient ischemic attacks, myocardial infarction within the previous 12 months, clinically significant arteriovenous (AV) conduction disturbances and/or arrythmias, unstable angina, history of heart failure, serum creatinine ≥200 mmol/L, serum potassium ≥5.5 mmol/L or ≤3.5 mmol/L, treatment with oral corticosteroids, concomitant use of agents that may affect BP except β-blockers and nitrates used in the treatment of stable angina, drug or alcohol abuse, pregnancy, breast feeding, and ineffective contraception.

Study design

After screening assessment, current antihypertensive medications, other than β-blockers and/or nitrates for patients with stable angina, were discontinued during a washout period of seven days (14 days for ACE inhibitor-treated subjects). A two- to four-week single-blind placebo run-in period was followed by a 16-week titration period and a 36-week maintenance period. At the end of the run-in period, subjects with a clinic SIDBP of 90 to 115 mm Hg and increased UAE were randomized to double-blind treatment with either losartan 50 mg or enalapril 5 mg o.d. in a 1:1 ratio, within each center. Four weeks later, patients receiving enalapril had their medication titrated up to 10 mg o.d. if their SIDBP was above 85 mm Hg, while those receiving losartan were maintained on 50 mg o.d. At week 8 of double-blind treatment, uncontrolled subjects (SIDBP > 85 mm Hg) of both groups had their medication doubled. At week 12, subjects with SIDBP > 85 mm Hg were given a combination of losartan or enalapril plus hydrochlorothiazide (HCTZ) 12.5 mg titrated to 25 mg to achieve a goal SIDBP of 85 mm Hg. Additional antihypertensive agents other than ACE inhibitors, AT_1 antagonists, and calcium channel blockers were then prescribed to achieve goal BP control. Early up-titration was permitted starting at week 4 for patients having SIDBP >105 mm Hg. At week 20 of double-blind therapy, subjects with SIDBP > 100 mm Hg were discontinued from the study. No planned modification was made in the subject's usual insulin or antihyperglycemic agent regimen throughout the study. In addition to the subject's continuous autosurveillance of their glycemia, chemical indices of control such as glycated hemoglobin (HbA_1C) and plasma glucose level were determined at regular intervals. The subject's diets were unchanged, with no sodium or protein restriction.

The antihypertensive effects of the study medications were evaluated by the results of the clinic BP measurements and ABPs. Clinic BP was measured at each visit between 7 and 10 a.m., 24 hours after medication ingestion (range 22 to 26 h), with a standard mercury sphygmomanometer and an appropriately sized cuff, and with Korotkoff phases I and V for the systolic and diastolic values, respectively. All readings were performed in triplicate in both sitting and standing positions after a five-minute rest period. Values were then averaged. Twenty-four–hour ambulatory BP was measured noninvasively at 20-minute intervals, between 7 a.m. and 10 p.m. (daytime) and at 30-minute intervals between 10 p.m. and 7 a.m. (nighttime), using the Spacelabs model 90207 (Spacelabs M.C., Redmond, WA, USA) oscillometric device. ABP monitoring occurred four times during the study: at the end of the placebo run-in period and after 12, 28 and 52 weeks of the double-blind period. Monitoring accuracy was validated against a conventional mercury sphygmomanometer using a t-tube connector [15]. The following quality-control criteria were established as standards for acceptability for each ABP monitoring report: (*1*) a minimum of 24 hours of data postdose, (*2*) a minimum of 50 total valid readings (80% of total readings), (*3*) a minimum of 22 total valid reading hours, and (*4*) no two consecutive invalid hours. Twenty-four–hour daytime and nighttime BP values were derived from the arithmetic mean of all the values recorded during the corresponding periods.

Laboratory evaluations, including hematology, chemistry, HbA_1C, lipid profile, and urinalysis, were performed during the baseline period and after 4, 12, 28, and 52 weeks of active treatment. Tolerability of study treat-

ment was assessed by monitoring of spontaneous reports of adverse experiences at each visit. UAE was measured on three consecutive days, utilizing timed overnight (8 to 12 hour) urine collections [16] during the baseline period and after 12, 28, and 52 weeks of active treatment. Analysis of albumin concentration was made by radioimmunoassay (Pharmacia Albumin RIA; Pharmacia Diagnostic AB, Uppsala, Sweden). The mean of the three samples was calculated. A urine culture was performed if subjects had symptoms of urinary tract infection. These subjects could be enrolled when their urine culture became negative. GFR was estimated, at the same time points as UAE, by the plasma disappearance of ^{51}CR-ethylenediaminetetraacetic acid (EDTA). The values were standardized for a body surface area of 1.73 m^2 [17].

Statistical analysis

Data from Lacourcière et al [7] and Chan et al [18] were used for the power calculation. The calculation was carried out on the variable defined as LOG_{10} (UAE at exit/UAE at baseline). Assuming 40 evaluable subjects in each group, a power of 80% and $\alpha = 0.05$ (two tailed) were calculated in order to detect a within-group change from baseline in log UAE of 0.20 or greater. With the same conditions and level of power, the detectable difference between groups for the change from baseline in log UAE is 0.27 or greater.

All analyses were performed using version 6.12 of the SAS statistical package. All statistical comparisons were based on two-sided tests. Statistical significance of treatment comparisons was declared if the probability value was less than or equal to 0.05 for each time point. Treatment differences were evaluated using analysis of variance (ANOVA) on the change from baseline considering an intention-to-treat approach. A two-way ANOVA model with treatment effect, center effect, and treatment-by-center interaction was used. The latter interaction term was removed if there was no evidence of significant interaction ($P > 0.10$). The qualitative and quantitative nature of the interaction involving treatment effects on UAE was tested by the method of Gail and Simon. Calculations of UAE and GFR were performed on log-transformed values and were presented as geometric means. Skewness in the distribution of the data was corrected by log transformation. The normality and homogeneity of the variances were verified by the Shapiro–Wilk test and the Levene test, respectively. Correlations between changes in BP and UAE were evaluated by Pearson's analysis. The safety of treatments was assessed by means of Fisher's exact test.

RESULTS

Study population

Of the 103 hypertensive type 2 diabetics who entered the double-blind treatment period, 92 (89.3%) completed the study, 46 subjects in each treatment group. There were 49 subjects in each group included in the intention-to-treat analysis for UAE at week 52. Six subjects were discontinued from the study in the losartan group and five in the enalapril group. Two subjects, one in each group, were withdrawn for uncontrolled hypertension, while one subject in the losartan group was discontinued as a result of urticaria and one in the enalapril group as a result of cough. Deviations from protocol account for the discontinuation of four subjects in the losartan group and two in the enalapril group. Another subject in the enalapril group voluntarily withdrew his consent. There were no deaths nor any cardiovascular events during the course of this study. Baseline gender distribution, race, age, and weight were similar in the two groups (Table 1). However, SIDBP at baseline was higher in the losartan group ($P = 0.025$), and the mean duration of diabetes was longer ($P = 0.039$) in the losartan group.

Table 1. Patient characteristics at baseline

	Losartan ($N = 52$)	Enalapril ($N = 51$)
Gender		
Male	39	44
Female	13	7
Mean age (SD)	59.2 (9.2)	57.8 (10.5)
Race (*N*)		
Caucasian	50	49
Oriental	1	2
Black	1	0
Mean sitting clinic BP *mm Hg* (SD)		
Systolic	162.3 (16.2)	157.7 (15.9)
Diastolic	97.2 (6.3)[a]	95.3 (4.8)
Weight *kg* (SD)	92.4 (17.2)	91.5 (19.8)
Mean duration of diabetes *years* (SD)	9.2 (7.5)[b]	12.6 (8.4)
Mean age at diabetes diagnosis *years* (SD)	49.7 (10.7)[c]	45 (10.6)
Mean UAE (geometric means) *μg/min*	64.1	73.9

[a]$P = 0.025$; [b]$P = 0.031$ vs. enalapril
[c]$P = 0.039$ vs. losartan

By the end of 52 weeks of active treatment, 20 out of 52 subjects in the losartan group received monotherapy (38.5%), and 31 out of 52 required combination therapy with HCTZ (59.6%), including 12 patients on triple therapy (23%). One patient randomized to losartan never received any treatment. In the enalapril-treated group, 25 out of 51 subjects received monotherapy (49%), and 26 out of 51 received combination therapy with HCTZ (51%), including five patients on triple therapy (5.8%). The mean dose (SD) of losartan was 86.3 (22.5) mg versus 16.0 (6.2) mg for enalapril. Mean doses of HCTZ were 23.0 (4.7) mg for the losartan group and 21.6 (5.7) mg for the enalapril group. Triple therapy consisted of losartan or enalapril plus HCTZ and β- or α_1-adrenoreceptor blockers. There were no significant differences in

Table 2. Ambulatory blood pressure responses (mean changes mm Hg)[a] to losartan and enalapril

	Losartan (N = 38)				Enalapril (N = 40)			
	Baseline	12 week	28 week	52 week	Baseline	12 week	28 week	52 week
Systolic								
24 hour	158.4	149.4 (−9.0)	142.8 (−15.7)	144.0 (−14.4)	153.7	143.4 (−10.3)	138.2 (−15.5)	138.3 (−15.4)
Day	162.8	154.0 (−8.8)	146.1 (−16.8)	147.2 (−15.6)	159.9	147.8 (−12.1)	142.6 (−17.3)	142.7 (−17.2)
Night	150.0[b]	143.9 (−6.1)	135.7 (−14.3)	136.4 (−13.6)	144.0	136.2 (−7.9)	130.9 (−13.1)	130.0 (−14.0)
Diastolic								
24 hour	89.7	84.8 (−5.0)	80.7 (−9.0)	81.2 (−8.6)	88.4	82.4 (−6.0)	79.7 (−8.7)	78.8 (−9.6)
Day	93.1	87.8 (−5.3)	83.3 (−9.8)	83.7 (−9.4)	92.7	86.2 (−6.6)	83.4 (−9.4)	82.6 (−10.1)
Night	83.9	79.0 (−5.0)	75.1 (−8.8)	74.9 (−9.0)	80.8	75.8 (−5.1)	73.7 (−7.1)	71.9 (−8.9)

Day from 7 AM to 10 PM; night, from 10 PM to 7 AM.
[a]Significantly different ($P \leq 0.002$) from baseline for all values
[b]$P < 0.05$ vs. enalapril

the use of combination therapy (P = 0.324) or in the use of other antihypertensive agents (P = 0.11) for the two groups of patients. ABP was performed on willing subjects only at five of the eight sites. Of the 89 subjects who entered the study at the five centers, 78 provided acceptable and evaluable ABP measurements, 38 subjects in the losartan group and 40 in the enalapril group. The three sites that did not participate in the ABP monitoring preferred to be exempt from carrying out this procedure on their patients, as they were not completely familiar with the technique. There was no significant change in body weight as well as in dietary protein consumption in the treatment groups during the trial.

Effects on blood pressure

After 52 weeks of active treatment, both losartan and enalapril administered alone or in combination with other agents significantly decreased ($P < 0.005$) clinic sitting systolic BP (SISBP) and SIDBP without a clear-cut difference between the two treatment groups. Indeed, losartan decreased sitting BP from 163.3 (16.2)/97.2 (6.3) mm Hg to 148.3 (17.1)/86.8 (9.6) mm Hg. On the other hand, enalapril decreased sitting BP from 157.7 (15.9)/95.3 (4.8) mm Hg to 145.5 (18.2)/84.4 (8.4) mm Hg. Mean changes in standing BP were similar to changes observed in the sitting position. The percentage of subjects who achieved goal SIDBP of ≤ 85 mm Hg at the end of the study was 49% in the losartan group and 51% in the enalapril group.

Baseline ABP levels were higher in the losartan group during each period, although statistical difference was demonstrated only for nighttime systolic BP (Table 2). Both losartan and enalapril administered alone or in combination with other antihypertensive agents were effective in reducing BP during 24-hour daytime and nighttime periods, without statistical differences between groups.

Effects on urinary albumin excretion

Of the 103 randomized subjects, 93 had baseline UAE 20 to 200 μg/min (microalbuminuria), and 10 subjects had UAE 200 to 300 μg/min. Three patients with microalbuminuria upon enrollment raised their UAE > 200 μg/min during the placebo run-in period. There was no statistically significant difference between groups in UAE at baseline. The mean decrease in UAE was significant by week 12 ($P \leq 0.027$) in both treatment groups and remained significantly lower ($P < 0.001$) than baseline values throughout the study period. Albuminuria (geometric means) decreased from 64.1 μg/min at baseline to 55.1, 36.8, and 41.5 μg/min after 12, 28, and 52 weeks, respectively, in the losartan group, and from 73.9 μg/min to 50.7, 39.4, and 33.5 μg/min for the same time intervals, in the enalapril group (Fig. 1). There was no significant difference between groups with respect to the change from baseline in LOG UAE after 12 and 28 weeks of treatment. At week 52, analyses showed a significant quantitative treatment-by-center interaction characterized by a variation in the magnitude of treatment differences from center to center. The difference between groups with respect to the change from baseline in LOG UAE is not significant when the interaction is taken into account and significant (P = 0.026) otherwise.

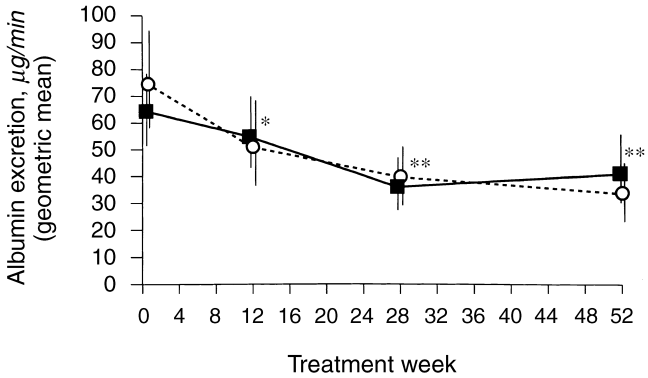

Fig. 1. Urine albumin excretion in the losartan (■) and enalapril (○) groups. Both treatments induced significant reductions from baseline in albuminuria at either 12, 28, or 52 weeks of therapy. Error bars represent 95% confidence intervals. *P = 0.027; **$P < 0.001$ vs. baseline.

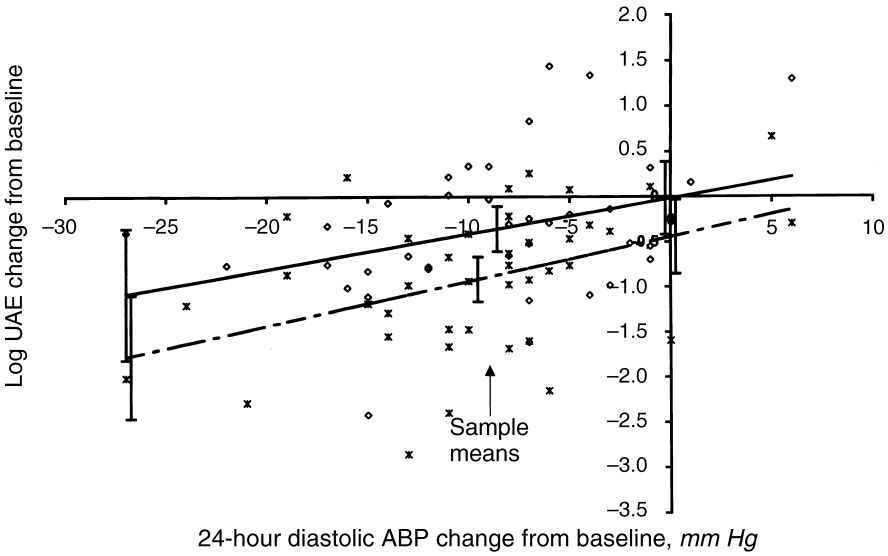

Fig. 2. Relationship between the changes in ambulatory blood pressure (ABP) and the changes in Log urinary albumin excretion (UAE) for losartan (N = 38; ◇) and enalapril (N = 40; ×) groups at week 52 is illustrated by the regression lines fitted to the data points of the scatter plot. Losartan (LOS) slope = 0.0395, Enalapril (ENA) slope = 0.0495. P = 0.69 between groups; r = 0.34, P = 0.039 for LOS; r = 0.42, P = 0.007 for ENA (error bars represent 95% confidence units on regression means).

The observed numeric difference should therefore be interpreted with caution.

Changes in LOG UAE correlated significantly ($P <$ 0.05) with changes in 24-hour systolic or diastolic ABP and nighttime systolic ABP (mm Hg) at week 52, for both treatment groups combined. Analyzing each group separately, the correlation obtained with changes in 24-hour systolic ABP was r = 0.31, P = 0.057 for losartan and r = 0.41, P = 0.008 for enalapril. A significant correlation ($P <$ 0.05) was obtained with changes in 24-hour diastolic ABP (Fig. 2) for each group (r = 0.34 for losartan and r = 0.42 for enalapril). With changes in nighttime systolic ABP, the correlation was r = 0.33, P = 0.045 for losartan and r = 0.31, P = 0.051 for enalapril. There were no statistical differences between the two treatment groups for any of the correlation coefficients obtained. Accordingly, there was no statistical difference in the linear relationship (regression slope) of the changes from baseline in Log UAE with respect to the changes from baseline in ambulatory BP at week 52 between the two groups. In contrast, the reduction in UAE was less well correlated with the reduction in clinic BP. A significant but weaker correlation (r = 0.23, P = 0.022) was seen with the change in SIDBP for both groups combined. Again, there was no statistical difference between treatments.

Effects on renal function

Baseline GFR (geometric mean) was nearly identical for the losartan (96.7 mL/min)- and enalapril (95.3 mL/min)-treated subjects (Fig. 3). A similar decline in GFR was observed with each treatment group starting at week 12. The overall decline of approximately 9% between baseline and week 52 was significant ($P \leq 0.001$) in both groups. The rate of decline plateaued by the end of treatment. The GFR fall during the first 12 weeks was positively related to the degree of 24-hour mean systolic (r = 0.24, $P <$ 0.05) and diastolic (r = 0.23, $P <$ 0.05) ABP decrements. However, the correlation was no longer significant after 28 and 52 weeks of treatment.

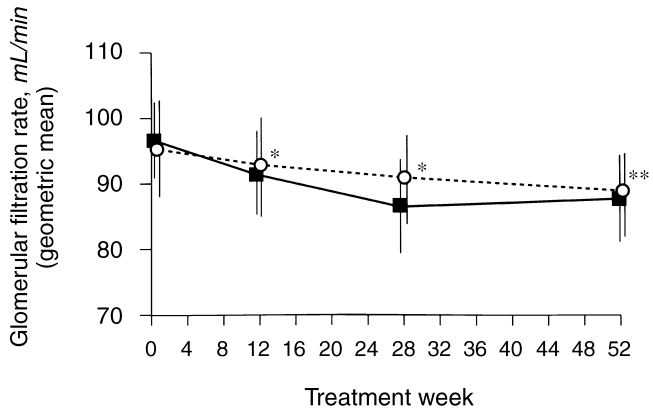

Fig. 3. Glomerular filtration rate (GFR) in the losartan (■) and enalapril (○) groups. Both treatments induced a significant reduction from baseline in GFR that plateaued after 52 weeks of therapy. Error bars represent 95% confidence intervals. *P = 0.006 for losartan; **$P <$ 0.001 for both losartan and enalapril versus baseline.

Clinical and laboratory safety

There were no significant differences between the losartan and enalapril treatment groups in the frequency of total clinical adverse experiences. However, treatment-related cough occurred significantly more frequently in those subjects taking enalapril (14% vs. 0%; P = 0.006). Two subjects (one with dyspnea and the other one with urticaria) in the losartan group and one subject with cough in the enalapril group discontinued the active treatment because of adverse clinical experiences.

There were no significant changes in serum glucose. The median value for HbA1c at baseline tended to be lower in the losartan group (0.077) compared with the enalapril group (0.085). There were slight but statistically significant ($P \leq 0.005$) increases in both groups at week 28. The change from baseline remained significant in the losartan group, but not in the enalapril group, at the end of the study. HbA1c median changes were +0.006 for losartan and +0.0025 for enalapril at week 52. Subjects in both groups remained within the range of fair diabetic control throughout the study as assessed by this laboratory parameter. There were statistically significant ($P < 0.05$) median changes at week 52 in total cholesterol (2.1 and 4.2% decrease within the losartan and enalapril group, respectively), triglycerides (11.3% decrease in the enalapril group), and low-density lipoproteins (6.5% decrease in the losartan group).

Of particular interest is the magnitude of changes in serum concentration of uric acid. Median changes of −22.0 μmol/L in the losartan group and of +12.0 μmol/L in the enalapril group ($P = 0.001$ enalapril vs. losartan) were observed after 12 weeks of monotherapy. After 52 weeks of active monotherapy or combination therapy, median changes of +2.0 μmol/L in the losartan group and of +30.0 μmol/L in the enalapril group ($P = 0.002$ enalapril vs. losartan) were observed.

DISCUSSION

The main purpose of this long-term double-blind study was to compare the effects of the AT_1 antagonist losartan to those of the ACE inhibitor enalapril on kidney function in hypertensive type 2 diabetics with early nephropathy. Our results indicate that UAE was significantly reduced with both losartan and enalapril treatments. The observed anti-albuminuric effect of losartan in this study is consistent with results from previous studies in which AT_1 antagonists reduced proteinuria [12–14].

Ambulatory BP monitoring is a well-established method that allows the determination of circadian variations of BP and the efficacy of antihypertensive treatment [19]. Moreover, ABP monitoring has been found to be more closely related to target organ damage [20, 21] and to cardiovascular mortality than clinic BP [22, 23]. Furthermore, clinic BP has been found to be less closely related to microalbuminuria than ABP in patients with essential hypertension [24] and in those with type 2 diabetes mellitus [25]. In the present study, there is a correlation between the decreases in ABP and UAE. These results are in agreement with those of Bauduceau et al, who have shown that ABP decrease is a valuable predictor of UAE decrease in hypertensive type 2 diabetics [26]. The significant correlation between ABP decrease and the changes in UAE may explain, at least in part, the apparently greater UAE decline in the enalapril group compared with the losartan group seen at week 52. However, taking into account individual changes in ABP, the linear relationship of the changes from baseline in Log UAE with respect to the changes from baseline in ABP was similar in both treatment groups.

Some center-to-center variation is to be expected in multicenter studies. In the present study, there is a treatment-by-center interaction whereby the treatment differences across centers vary significantly in magnitude. Possible sources of interaction include factors such as the duration of diabetes at entry and age at the time of diagnosis of the disease. As discussed earlier, another factor that can explain differences in the effect of the two treatments on UAE is the change in ABP. The extent to which decreases in ABP and UAE are correlated suggests that the antiproteinuric effects of AT_1 antagonists and ACE inhibitors may also be mediated by other mechanisms such as distinct renal effects involving microcirculation changes and tissue remodeling. The concept that the renal effects of AT_1 antagonists and ACE inhibitors are mediated by interference on the RAAS rather than the kinin-kallikrein system is supported by the results of other studies showing similar systemic and hemodynamic effects of these agents [12, 27].

Another important aim of the present study was to compare the effects of AT_1 antagonists and of ACE inhibitors on GFR in hypertensive type 2 diabetics. Our data provide the first long-term evidence, to our knowledge, that AT_1 antagonists have similar effects to ACE inhibitors on the rate of decline in GFR. The decrease in GFR during the first three to seven months after the start of antihypertensive therapy is consistent with previous studies using ACE inhibitors [6, 28] and is most likely caused by an hemodynamic-induced redistribution of kidney blood flow, secondary to the abrupt lowering of systemic BP. Further studies are, however, required to examine whether longer term intensive control of BP with AT_1 antagonists can stabilize the decline in GFR and delay the loss of renal function, as previously demonstrated with ACE inhibitor therapy [28, 29].

Despite the beneficial clinical effects related to interference on the RAAS, the high prevalence of cough associated with ACE inhibitors has reduced their general acceptability. In the present study, cough was not induced by losartan, while the incidence was 14% in the enalapril treated patients. These findings are in agreement with a previous study showing that the cough observed with ACE inhibitors does not occur with the AT_1 antagonists [11]. Thus, treatment with these new agents may offer a valuable alternative for patients with ACE inhibitor-related cough.

Although raised serum uric acid concentration is commonly encountered in hypertensive patients, contradictory evidence exists as to whether serum uric acid is a cardiovascular risk factor [30, 31]. As reported earlier,

in the present study uric acid remained stable in the losartan treated patients in spite of the fact that a greater number of subjects required concomitant administration of HCTZ [32, 33]. The increased in uric acid excretion mediated by losartan may theoretically correct the relatively diminished uric acid excretion rate in hypertensive patients [34]. However, the beneficial influence of losartan on the frequency of gout attacks in hypertensive patients treated or not with diuretics versus ACE inhibitor or other antihypertensive treatments remains to be established in prospective studies.

In conclusion, in hypertensive type 2 diabetics with early nephropathy, treatment for 12 months with losartan or enalapril reduced UAE significantly and showed a similar response in UAE reduction with respect to the decrease in ABP. Long-term studies are required to examine whether the antiproteinuric effects of AT_1 antagonists confer significant renoprotection.

ACKNOWLEDGMENTS

This study was supported by a grant from Merck Frosst Canada & Co. We are grateful to Dr. Jean Campeau (CHRA, Thetford Mines, Québec, Canada) and to Dr. Yves Pesant (Hôtel Dieu de St-Jérôme, Québec, Canada) for their participation in the study. Special thanks are extended to Mr. Daniel Morissette for performing the statistical analysis and to Mr. Luc Poirier and Mr. Jean Lefebvre for their pharmacologic advice. The results of the present study were presented in abstract form at the European Society of Hypertension meeting, Milan, June 1999.

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